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Prof. Derek Abbott, Editor-in-Chief
IEEE Access
Institute of Electrical and Electronics Engineers (IEEE)

Subject: Submission of Original Research Manuscript to *IEEE Access*
Title: *Integrating Multimodal Earth Observations and Hydrological Regularization for Spatiotemporal Wetland Water Mapping in the Indus Delta*

Dear Prof. Abbott and Members of the Editorial Board,

Please find enclosed our original research manuscript entitled **"Integrating Multimodal Earth Observations and Hydrological Regularization for Spatiotemporal Wetland Water Mapping in the Indus Delta"** for consideration for publication in *IEEE Access*.

Scope and Multidisciplinary Relevance to *IEEE Access*: This work sits at the intersection of **Geoscience and Remote Sensing**, **Computational Intelligence / Deep Learning**, and **Environmental Systems Engineering**. Mapping dynamic surface water across data-scarce alluvial and deltaic basins is hindered by severe spatial resolution disparities between multi-sensor satellite observations and coarse hydroclimatological reanalysis products, as well as the tendency of purely data-driven deep neural networks to violate physical catchment mass conservation and overfit static historical occurrence priors.

Key Technical Contributions and Verified Findings:

1. **Multimodal Flood Physics-Regularized Transformer (MFPIT):** We propose a compact (6.58M-parameter; $6,577,318$ exact parameters) hybrid CNN–Transformer architecture that fuses 12-month multi-sensor Earth observation sequences (Sentinel-2 MSI, MODIS, Sentinel-1 SAR, and SRTM topography) with regional hydroclimatological fluxes (CHIRPS precipitation and TerraClimate evapotranspiration, soil moisture, and surface runoff) regularized by a dimensionless standardized catchment mass-balance loss ($\mathcal{L}_{\text{phys}}, \lambda_{\text{phys}} = 0.1$) and an inverse-rainfall weighted temporal smoothness constraint ($\mathcal{L}_{\text{temp}}, \lambda_{\text{temp}} = 0.1$).

2. **Forensic Target Leakage Audit (JRC = 0) and 23-Year Spatiotemporal Benchmark:** Over a 23-year monthly archive (2001–2023; 279×502 basin-scale grid at 1 km resolution across the Lower Indus Basin and Delta; $1,862$ training sequences and 196 validation/test sequences), we demonstrate that retaining static JRC occurrence priors induces trivial feature–target leakage (IoU = 1.0000). Under our strictly leak-free JRC = 0 benchmark with validation-calibrated decision thresholds, MFPIT ($B = 8$) achieves state-of-the-art segmentation (Test IoU = 0.7102, F1 = 0.8306 at $\tau^* = 0.330$), probabilistic ranking (PR-AUC = 0.9141, ROC-AUC = 0.9296), and probability calibration (ECE = 0.0150, Brier = 0.1057), significantly outperforming Random Forest ($p = 9.59 \times 10^{-11}$), GBDT ($p = 4.59 \times 10^{-9}$), U-Net ($p = 7.68 \times 10^{-21}$), and CNN-LSTM ($p = 1.38 \times 10^{-29}$).

3. **Out-of-Distribution Mega-Flood Resilience and Dynamic 2022–2023 Flood Mask Evaluation:** Joint physical and temporal regularization reduces test-set catchment mass-balance residuals by 81.7%–89.6% and temporal jitter by 86.7%, yielding a patch water-fraction R^2 of 0.9564 (MAE = 0.0458). During the catastrophic August–September 2022 Pakistan mega-flood ($+2.20\sigma$ runoff shock), unconstrained training ($\lambda_{\text{phys}} = 0.0$) collapses to IoU = 0.4967 ($R^2 = 0.2959$), whereas physics-regularized MFPIT maintains IoU = 0.6962 (+19.95 percentage points) and $R^2 = 0.9382$, while also ranking first across all models on a dynamic 2022–2023 monsoon inundation reference mask (IoU = 0.6456).

Open Science and Ethics Declarations: All code, Google Earth Engine extraction scripts, and evaluation pipelines are publicly available at `https://github.com/mirzamuzzamilbaig/MFPIT`. This manuscript is original, has not been published previously, and is not currently under consideration by any other journal or conference. The author declares no competing financial or personal interests.

Thank you very much for your time and consideration.

Sincerely,

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